Masud Pervez

Senior Data Scientist | ML Engineer | Statistician



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Summary

Experienced and forward-thinking Data Scientist with over eight years of experience and a robust academic foundation in Statistics. Proven track record in building and deploying production-grade fraud detection systems, predictive models, and scalable data pipelines. Hands-on expertise in Spark, Scala, and Python for end-to-end model development, deployment, and monitoring. Skilled in hypothesis testing, advanced analytics, and cost-based optimization to enhance fraud detection precision while minimizing financial losses. Strong collaborator with cross-functional teams, driving data-driven decisions aligned with business objectives.

Work Experience

Senior Data Scientist

Skandinaviska Enskilda Banken (SEB) | Stockholm, Sweden

June 2021 - Present

- Led the design and deployment of real-time machine learning systems detecting fraudulent transactions with over 60% recall while maintaining low false positive rates.
- Developed a cost-based threshold tuning framework to balance fraud loss and customer experience, optimizing overall fraud detection efficiency.
- Built modular Spark pipelines for data loading, feature computation, model inference, and metric computation on Hadoop clusters.
- Perform Data and Model Governance.
- Collaborated with engineers and cross-functional teams to deploy ML systems into production using via microservices architecture.
- Monitored model drift and performance, recommending model recalibration and retraining strategies.

Data Scientist

Bonnier News | Stockholm, Sweden

Sep 2018 - May 2021

- Developed churn, customer reactivation, and Customer Lifetime Value models to improve retention and marketing ROI.
- Built a personalized recommendation system to predict next-best magazine purchases and increase engagement.
- Increased profitability by 16% through a Newsvendor optimization model for distribution.
- Deployed and experimented with scalable ML solutions in GCP.
- Delivered data-driven insights across 40+ products and campaigns through A/B testing, customer retention analysis, and stakeholder dashboards, enabling better decision-making.

Core Competencies

- Fraud Detection & Risk Modeling
- Predictive Machine Learning & Cost Optimization
- Designed and Implement ML Systems
- Model Deployment and Governance
- Processed large-scale datasets & performed ML Engineering using Spark and distributed systems
- Python (Scikit-Learn, Flask, Pandas, PySpark)
- Exploratory Data Analysis (EDA) & Data Visualization
- Collaboration & Stakeholder Communication
- Analytical & Leadership skills

Academic History

M.Sc. in Mathematical Statistics

Division of Mathematical statistics, Stockholm University, Sweden

2008 - 2015

M.Sc. in Applied Statistics

Institute of Statistical Research and Training (I.S.R.T.), University of Dhaka. Bangladesh

2004 - 2006

B.Sc. in Applied Statistics

Institute of Statistical Research and Training (I.S.R.T.), University of Dhaka. Bangladesh

1998 - 2004

Work Experience

Data Scientist | Statistician

Tobii Pro | Stockholm, Sweden

May 2017 – Sep 2018

- Preprocessed and analyzed eye-tracking survey data for advanced analytics using Python and R.
- Applied statistical and ML methods (PCA, logistic regression, decision trees) for pattern discovery.
- Developed visual dashboards and reports to present insights to stakeholders.

Planner and Data Integrity Analyst

Marks and Spencer | Dhaka, Bangladesh

Oct 2007 - Aug 2008

- Analyzed supply chain data in various business aspects for the company.
- Prepared periodic (daily, monthly, yearly) business reports and KPI's based on shipment data.
- Investigated supplier's and freight forwarder's performance on a regular basis.
- Arranged training in order to improve supplier's performances
- Planned shipment of the container including goods, moving from origin to destination.

Master's degree Project (Statistician)

Clinical Cancer Epidemiology, Karolinska Institutet | Stockholm, Sweden

Sep 2012 - Dec 2013

- Analyzed survey questionnaire reported by gynecological cancer survivors or patients who had experienced cancer in their lives and are treated with radiation therapy and surgery treatments.
- Study to identify whether the measured questions interpret the same symptom based on the given answers or they describe in different ways.
- Compare different hierarchical clustering results using a cluster comparison method.
- Performed factor analysis to extract the underlying factors associated with the symptoms.

Emerging Interests

- Large Language Models (LLMs)
- Generative AI & recommender systems
- Retrieval-Augmented Generation (RAG)

Kev Achievements

• Designed and Implemented Machine Learning System

that predicted fraudulent and non-fraudulent transactions in real-time with more than 70% recall.

• Optimized Profits by Implementing the Newsvendor Model in Distribution

that resulted in 16% profit growth for the company.

• Collaborated with diverse business and technical teams to evaluate and assess external vendors (RFP), enhancing the bank's financial crime prevention capabilities.

Technical Skills

- Programming: Python, Scala, R
- Libraries/Frameworks: Scikit-Learn, TensorFlow, PyTorch, Flask, Pandas, NumPy
- Big Data: Spark, Hadoop, HDFS, Hive
- Cloud & Deployment: GCP, OpenShift, Docker, Airflow, CI/CD
- Database: SQL, PostgreSQL
- Other: Git, Web Scraping, API Development, Data Visualization.
- AI Agents, Langchain, Langgraph

Machine Learning Engineering & MLOps

- Built and deployed ML models using microservices architecture
- CI/CD pipelines for model deployment
- Model monitoring, drift detection, retraining strategies

Languages:

- English: Fluent
- Swedish: Intermediate

Reference

Reference can be provided upon request.

I hereby declare that the above information given herewith is true, complete, correct and up to date to the best of my knowledge and belief.

Masud Pervez